Multiscale Spatial–Spectral Invertible Compensation Network for Hyperspectral Remote Sensing Image Denoising

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***Abstract*— Hyperspectral image (HSI) has fine spectral resolution and abundant spatial information to detect subtle differences between targets. However, it is heavily contaminated with noise due to sensor design and atmospheric radiative transfer, resulting in spectral shifts and spatial discontinuities. Current denoising methods usually establish constraints directly on the ground truth and denoised image, lacking supervision of intermediate parameters of the network, resulting in insufficient model constraints and poor convergence. In addition, existing methods do not consider spatial–spectral compensation, so the denoising results have obvious spatial–spectral distortion. To this end, we propose a novel multiscale spatial–spectral invertible compensation network (MSIC-Net) for HSI denoising. The method constructs an invertible spatial–spectral compensation (ISSC) module, which supervises intermediate features through inverse constraints, realizes the circulation of multiscale information, and improves the stability of the model. At the same time, we also introduce style transfer for spatial–spectral compensation, which uses its superior fine feature control ability to precisely compensate for the lost spatial and spectral detail features. The method is extensively validated experimentally and categorically on simulated and real datasets. The experimental results show that MSIC-Net outperforms other state-of-the-art denoising methods in quantitative and qualitative evaluations.**

***Index Terms*— Convolutional neural networks (CNNs), hyperspectral image (HSI) denoising, spatial–spectral invertible compensation.**

I. Introduction

HYPERSPECTRAL remote sensing images have hundreds or even thousands of spectral bands, with fine spectral resolution, which can recognize subtle differences between features. It shows great application value and plays an important role in fine agriculture [1], geological exploration [2], and environmental monitoring [3]. However, hyperspectral images (HSIs) are inevitably interfered by noise in the process of imaging and transmission, and the complex noise types (Gaussian noise, impulse noise, stripes, and deadlines) reduce the resolution of the images and increase the difficulty of subsequent application research. To cope with the interference of complex noise, a denoising process that removes the noise and maximizes the retention of useful information is necessary.

Currently, a large number of HSI denoising algorithms have emerged, including two main categories: model-driven and data-driven. Model-driven methods are based on matrix decomposition theory, such as nonlocal similarity, sparse, and low-rank prior. By designing regularization constraints, convex optimization solving is used to achieve noise removal. Typical methods include BM4D [4], MTGSNMF [5], RhyDe [6], and LRMR [7]. These methods can effectively use the intrinsic structural information of HSIs to improve the denoising effect without relying on a large number of training datasets and have strong interpretability. Unfortunately, these methods often necessitate manual parameter adjustments to optimize the denoising results, which may entail substantial trial and error. Furthermore, these a priori theories typically encapsulate only a subset of the features of HSIs and are restricted to addressing a specific type of noise, thereby constraining their practical applicability in diverse denoising scenarios.

Compared to model-driven methods, data-driven methods do not rely on specific model assumptions but make the problem of HSI denoising more flexible through mapping learning. Such methods are highly adaptive and generalizable and can better handle different types of noise and complex scenes. Convolutional neural networks (CNNs) [8] have been widely used for their advantages in removing complex mixed noises. Typical methods include HSID-CNN [9], SSGN [10], LR-Net [11], and so on, which can consider both the spatial and spectral information of HSI to achieve satisfactory denoising results. In recent years, the introduction of novel network structures, such as transformers [12], generative adversarial networks [13], and diffusion models [14], has also provided new perspectives for HSI denoising.

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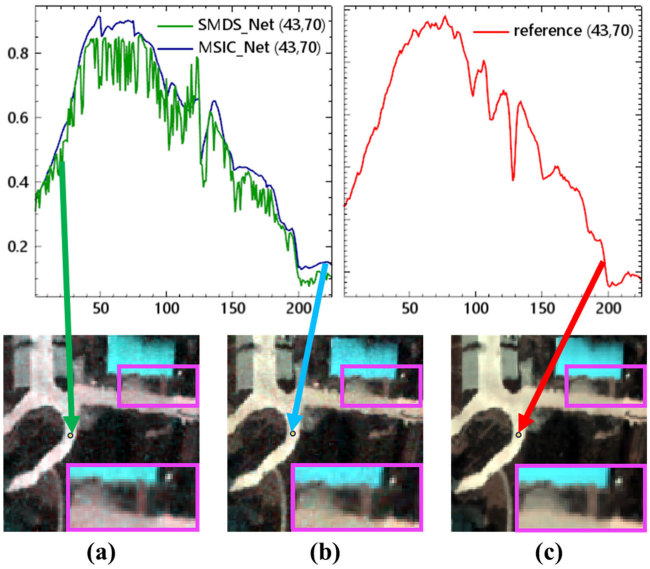


Fig. 1. Spectral profiles and spatial comparisons of HSI denoising. (a) SMDS-Net. (b) MSIC-Net. (c) Clean HSI.

However, current HSI denoising methods still face some challenges: 1) current data-driven denoising methods usually establish loss constraints directly on the ground truth and denoised images, as well as a lack of supervision on the network process parameters, which may lead to a decrease in model stability and more epochs are needed to obtain satisfactory denoising results, and 2) at present, despite the existence of many algorithms for HSI spatial and spectral feature extraction, there are still problems such as blurring, artifacts, and large interband differences in the denoising results (as shown in Fig. 1). In addition to the algorithm design, these problems arise from the fact that there is no compensation for features lost during the extraction process, resulting in noticeable spatial–spectral distortion in the denoising results. Therefore, maintaining the high quality of spatial and spectral information of HSI, while improving the stability of the model, is of great significance for the noise removal task.

To solve the above denoising problem of HSIs effectively, we design a novel multiscale spatial–spectral invertible compensation network (MSIC-Net). In MSIC-Net, we exploit the spatial and spectral redundancy inherent in HSIs to construct a spatial–spectral feature integration (SSFI) module at the first layer (l_1). This module employs 3-D convolution to extract highly structured features from the images. Furthermore, we introduce an efficient spatial and spectral attention mechanism to mitigate information redundancy and facilitate representative feature learning. This enhancement bolsters the accuracy and reliability of subsequent feature analysis. To address problem 1), we design the invertible spatial–spectral compensation (ISSC) module at other layers (l_2–l_4). This module supervises intermediate features through inverse constraints and incorporates multiconnection loss enhancement constraints to facilitate information flow and improve model stability. For problem 2), we construct a feature compensation block (FC-block) in the ISSC module, which introduces style transfer, capitalizes on its high controllability to refine the residual noise, and eliminates the potential factors of structured feature changes. It greatly reduces the information loss and realizes the spatial–spectral feature compensation. Our main contributions can be encapsulated as follows.

1) A novel MSIC-Net for HSI denoising is proposed, which enhances the network constraints and improves the convergence rate through an invertible multiscale architecture. Furthermore, spatial–spectral compensation is used for the first time for HSI denoising to achieve accurate HSI recovery.
2) We propose the ISSC module, which accurately supervises the network through inverse constraints and significantly improves the stability of the model. In addition, to achieve loss information compensation, we introduce style transfer to improve the quality of the extracted noise through random noise injection and multiscale training strategies.
3) Extensive experiments are executed to achieve comparable denoising results compared to other methods.

The remaining work is organized as follows. In Section II, we review the related HSI denoising methods. Section III describes our proposed MSIC-Net in detail, and the experimental results are demonstrated and analyzed in Section IV. Finally, conclusions are presented in Section V.

II. Related Work

In this section, we introduce the two mainstream methods of current HSI denoising: model- and data-driven methods in Sections II-A and II-B, respectively.

A. Model-Driven Methods

Based on the spectral correlation, nonnegativity, and spatial similarity of HSIs, the model-driven methods apply matrix decomposition (nonlocal similarity, sparse representation, and low-rank approximation) to build a regularized constrained model, and the denoising is achieved by using a convex optimization solution. Drawing upon the concept of nonlocal similarity, Dabov et al. [15] used nonlocal mean filtering to group similar 2-D images into 3-D arrays for noise removal via collaborative filtering. Maggioni et al. [4] enhanced denoising by grouping similar voxel cubes into a 4-D group and applying a 4-D transform. Despite the significant role of nonlocal similarity in HSI denoising, it ignores the spectral correlation of the image.

The sparse representation decomposes the image into dictionary atoms and coefficient matrices, and the sparse prior model is constructed to realize denoising considering the sparse differences between noisy signals and clear features. Qian and Ye [16] and Lu et al. [17] grouped similar pixels of sparse coding and denoised spatial and spectral information by constructing suitable dictionaries. Yang et al. [18] integrated spatial denoising and spectral decomposition by dictionary learning, while Lei et al. [5] preserved spatial information and spectral structure by sparse recovery and multitask learning.

The low-rank approximation views the corrupted data as low-dimensional data plus noise signal, and denoising is

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achieved by constructing a low-rank matrix to approximate the observed data. Zhang et al. [7] used dictionary ordering to arrange HSI patches into a 2-D matrix to eliminate mixed noise. He et al. [19] adaptively selected low-rank matrices according to the variation of noise intensity in different frequency bands to realize the denoising. Subsequently, considering the richness of information in the spectral dimension of HSI, He et al. [20], Xue et al. [21], Cao et al. [22], and Zhuang and Ng [23] simultaneously explore the low-rank structure of spectral and spatial dimensions to capture the features, which improves the HSI denoising performance. Vuong et al. [24] and Liu et al. [25] theoretically approximate the rank to automatically capture the feature of the HSI, which guarantees the spatial and spectral structure recovery.

Tensor decomposition methods that treat HSI as a higher order tensor and combine it with low-rank approximation are crucial in HSI denoising, including Tucker [26], CANDECOMP/PARAFAC (CP) [27], and tensor singular value [28] decompositions. These methods excel in denoising and inspire further improvements. Renard et al. [29] and Liu et al. [30] used multilinear algebra and PARAFAC decomposition to extract the spatial–spectral feature subspace and applied low-rank approximation for denoising. Guo et al. [31] and Xue et al. [32] considered the spatial–spectral correlation to form a low-rank tensor group and combined it with the rank estimation of CP decomposition to reconstruct HSI. Fan et al. [33] further considered the spatial–spectral smoothing of HSI and introduced total variation (TV) and group sparsity to capture the local correlations. Wang et al. [34] then used low-rank tensor Tucker decomposition to characterize the global correlation to achieve denoising.

B. Data-Driven Methods

Unlike model-driven methods, data-driven methods use a large amount of data to learn nonlinear mapping from degraded images to clean images. With its adaptability to complex noise environments, this method demonstrates significant promise in HSI denoising.

CNNs have been widely used in image restoration, and Xie and Li [35] first applied CNNs to HSI denoising using a trainable deep learning model. However, this band-by-band denoising method cannot retain the spectral features well, while the high-dimensional nature of HSI requires the combination of spatial–spectral strategies with deep learning, Xie and Li [35] combined spectral difference learning with the key band selection method based on residual learning to achieve spatial noise removal while finely preserving spectral reflectance. On the other hand, Chang et al. [36] modeled the spatial–spectral correlation of HSI using dilated convolution and multichannel filters to accelerate the training. Yuan et al. [9] and Maffei et al. [37] directly used the spatial and spectral features of HSI as the inputs of the network for estimating its noise, which enabled the network to make a better trade-off between denoising performance and detail preservation. To better construct the 3-D spatial–spectral correlation of HSI, Dong et al. [38] and Fu et al. [39] used an improved 3-D U-net to encode the intrinsic features for HSI denoising. Wei et al. [40] and Pan et al. [41] designed 3-D quasi-recurrent neural networks to capture spatial–spectral correlation to improve denoising performance. Liu and Lee [42] and Shi et al. [43] used 3-D dilated convolution and 3-D discriminative attention to extract features to effectively extract contextual information for HSI.

Emerging network architectures such as transformers, generative adversarial networks, and diffusion models have been increasingly applied to denoising. Li et al. [44], Yu et al. [45], and Tan et al. [46] used transformer architecture to explore the intrinsic features of HSI by designing unique spatial and spectral self-attention to capture interpixel and interspectral similarities. Fu et al. [47] combined transformer with recurrent neural networks to explore the global spectral correlation and nonlocal spatial self-similarity of HSI, achieving satisfactory denoising results. Zhang et al. [13] embedded the designed spatial–spectral module as a generator into the U-net structure and designed a multiscale feature fusion module as a discriminator to achieve denoising through adversarial learning. Luo et al. [48] proposed an unsupervised adaptive learning hyperspectral denoising network based on transfer learning, which explores the prior specific to real HSI and achieves satisfactory denoising results. Miao et al. [49] performed HSI restoration based on the robustness of the diffusion models to noise without any additional training data. Zeng et al. [50] introduced a self-supervised generative diffusion network to fuse spectral decomposition and conditional abundance generation to achieve satisfactory HSI denoising results. These innovative network structures offer new perspectives and solutions to HSI denoising, demonstrating the potential for further advancements.

In addition, the method of combining neural networks with a priori constraints (sparse and low-rank tensor) [11], [51], [52], [53], [54], [55], [56], [57] has been recognized as an effective denoising strategy that improves the interpretability of the model.

III. Proposed Method

The HSI $\mathcal{Y} \in \mathbb{R}^{I \times J \times K}$ obtained by the hyperspectral sensor is a 3-D tensor, where I and J denote the spatial dimensions and K denotes the number of spectral bands. The degradation model interfered with by mixed noise during the imaging process is as follows:

$$\mathcal{Y} = \mathcal{X} + \mathcal{N} \tag{1}$$

where $\mathcal{X} \in \mathbb{R}^{I \times J \times K}$ represents the clean HSI and $\mathcal{N} \in \mathbb{R}^{I \times J \times K}$ is the mixed noise. The aim of denoising is to recover the clear image from the degraded image disturbed by noise.

A. Multiscale Spatial–Spectral Invertible Compensation Network

The network structure of the MSIC-Net is shown in Fig. 2. MSIC-Net is a multiscale network consisting of four symmetric encoder–decoder pairs, with convolution and deconvolution capabilities. We implement downsampling and upsampling in the encoder and decoder using a step factor of 2 and add

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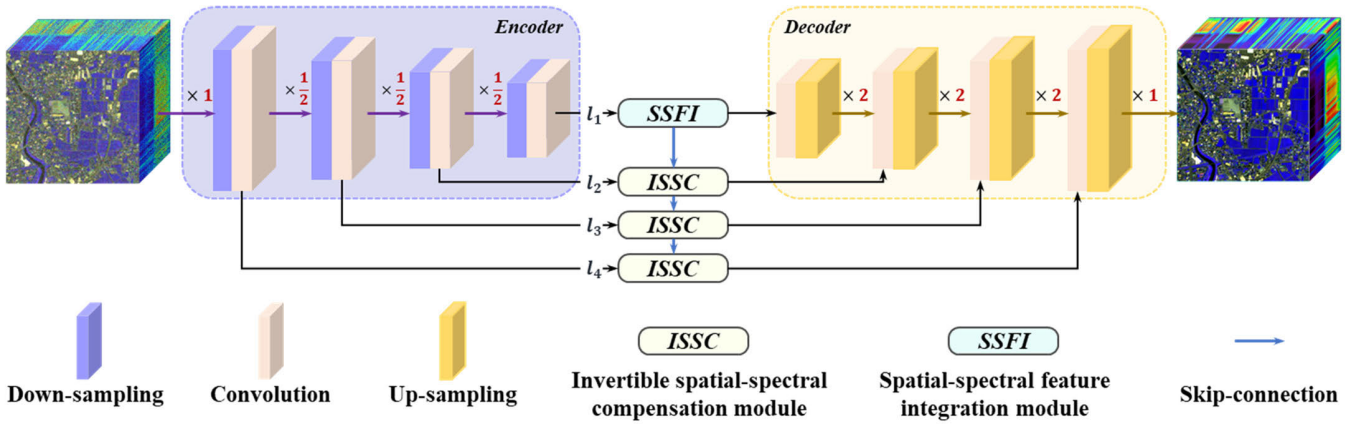


Fig. 2. Network structure of MSIC-Net. The frame is composed of a downsampling convolutional for the encoder (purple) and an upsampling convolutional for the decoder (yellow). The SSFI module is added to the first layer (l_1), and the ISSC module is added to the other three layers (l_2–l_4). In addition, skip connection is added to interoperate multiscale feature information.

skip connections between two neighboring layers to convey feature information at different scales. Relying on the advantage of smaller receptive fields to characterize geometrically detailed information, we designed the SSFI module in l_1 of the network. To reduce spatial and spectral redundancy, accentuate salient features, and accomplish the preliminary integration of structured features in HSIs, this module employs 3-D convolution for feature extraction. Further incorporation of spatial and spectral attention mechanisms is undertaken to optimize features. Although the small-scale network has a high resolution, the semantic information characterization ability is weak. To effectively use the spatial and spectral features captured by the SSFI module and reconstruct clean HSIs with high fidelity, we design the ISSC module in l_2–l_4, which leverages the nondestructiveness advantage of inverse constraints to reduce the loss of information, ensuring excellent stability of the model when dealing with complex noise. The accompanying FC-block relies on the highly controllable advantage of style transfer detail generation, compensating for feature detail from different receptive fields to use the semantic information in larger image areas, achieving spatial–spectral high fidelity.

B. Spatial–Spectral Features Integration Module (l_1)

In HSIs, the redundancy of spatial and spectral information is significant, mainly due to the correlation between neighboring pixels in spatial and the correlation between spectral channels. To efficiently extract features and reduce redundancy, the SSFI module is designed to integrate spatial and spectral information and highlight salient features. Its structure is shown in Fig. 3, which is a dual-branch structure capturing the spatial and spectral distribution characteristics of the features. Each branch consists of a 3-D convolutional layer for feature extraction and its attention to feature optimization. Finally, the information from the two branches is integrated using a $1 \times 1 \times 1$ convolution to obtain highly structured features.

Specifically, the feature extraction block (FE-block) uses 3-D convolution with different convolution kernels to capture the multiscale features from spatial and spectral distribution

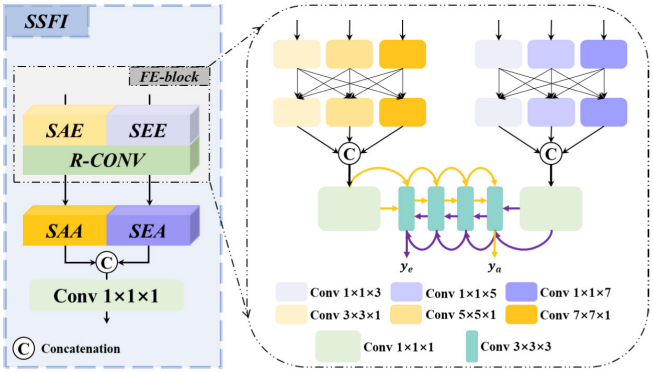


Fig. 3. SSFI module and its embedded **FE-block**.

characteristics to learn local and global salient feature representations. Among them, the spatial extraction unit (SAE unit) and spectral extraction unit (SEE unit) extract spatial and spectral distribution features, respectively. Afterward, the extracted features are sequentially subjected to recurrent 3-D convolution unit (R-CONV unit) to obtain the feature map y_a with spatial characteristics and y_e with spectral characteristics. The R-CONV unit not only captures the spectral and spatial intercorrelation in the input data but also reduces the computational cost of the model by relying on the advantage that the same layer can share weights. Immediately after, based on the lightweight spatial and channel reconstruction convolution (SCConv) [58], we introduced spatial and spectral attention units (SAA unit and SEA unit) (shown in Fig. 4) to reconstruct the features, reduce the spatial and spectral redundancy and accentuate the salient feature.

The SAA unit is used to assess the contribution of feature map y_a with spatial characteristics using learnable weight factors to achieve the separation of salient and irrelevant feature maps and finally to obtain a refined feature map $\hat{x}_a$ through cross-reconstruction. As shown in Fig. 4(a), the feature $y_a \in \mathbb{R}^{N\times K\times I\times J}$ is first normalized and the contribution of different spatial features is evaluated by the scaling factor $\gamma \in \mathbb{R}^K$ obtained by group normalization (GN) [59], where a larger γ indicates a richer spatial feature. The scaling

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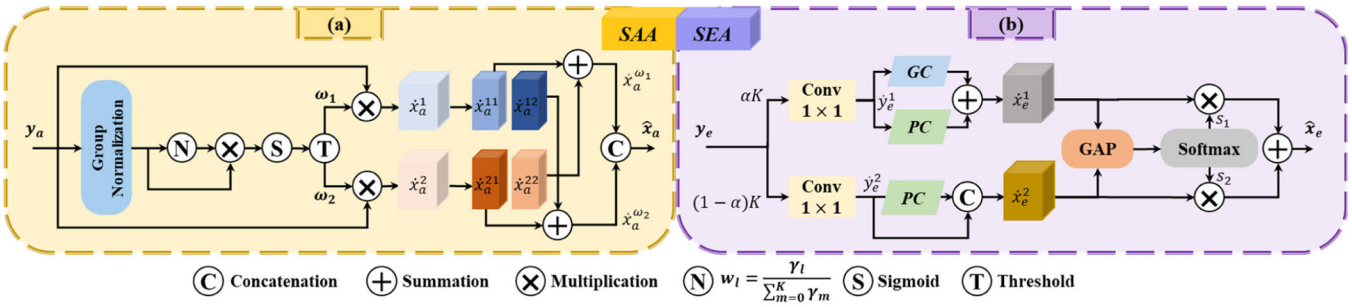


Fig. 4. (a) SAA unit. (b) SEA unit.

factor γ obtains the reweighted normalized weight $\mathcal{W}_\gamma \in \mathbb{R}^K$ through (2), which is immediately mapped to (0, 1) by the sigmoid function, and finally, the salient weight factor ω_1 and the irrelevant weight factor ω_2 are obtained through gating. The whole process of obtaining ω can be expressed as (3)

$$\mathcal{W}_\gamma = \{w_l\} = \frac{\gamma_l}{\sum_{m=0}^{K} \gamma_m} \tag{2}$$

$$\omega = \text{Gate}\left(\text{Sigmoid}\left(\mathcal{W}_\gamma\left(\gamma \frac{y_a - \bar{y}_a}{\sqrt{\sigma^2 + \varepsilon}} + \beta\right)\right)\right) \tag{3}$$

where N is the batch; K is the number of channels; I and J are the height and width of the spatial, respectively; $\bar{y}_a$ and σ are the mean and standard deviation of y_a, respectively; ε is a small positive; and γ and β are trainable affine transformations. Finally, the learned weight factors ω_1 and ω_2 are multiplied by the spatial feature y_a to obtain the salient feature $\dot{x}_a^1$ and the irrelevant feature $\dot{x}_a^2$. To make full use of the small amount of residual information in $\dot{x}_a^2$, the cross-reconstruction operation is used to enhance the flow of information to obtain the spatially informative and refined feature map $\hat{x}_a$. The whole process of the cross-reconstruction operation can be represented as

$$\begin{cases} y_a \otimes \omega_1 = \dot{x}_a^1 \\ y_a \otimes \omega_2 = \dot{x}_a^2 \\ \dot{x}_a^{11} \otimes \dot{x}_a^{22} = \dot{x}_a^{\omega_1} \\ \dot{x}_a^{21} \otimes \dot{x}_a^{12} = \dot{x}_a^{\omega_2} \\ \dot{x}_a^{\omega_1} \cup \dot{x}_a^{\omega_2} = \hat{x}_a \end{cases} \tag{4}$$

where $\otimes$ is the multiplication, $\otimes$ is the summation, and $\cup$ represents the concatenation.

On the other hand, the SEA unit reconstructs the feature map y_e with spectral characteristics along the channel dimension, realizes the circulation of features, reduces the redundancy of channel, and obtains the refined feature map $\hat{x}_e$. As shown in Fig. 4(b), the feature $y_e \in \mathbb{R}^{K \times I \times J}$ is partitioned into two parts by channel: one part has the channel number of αK and the other part has the channel number of $(1-\alpha)K$, where $\alpha \in [0, 1]$. Afterward, the channel numbers of the two groups of features are compressed using 1×1 convolution to obtain the salient feature $\dot{y}_e^1$ and the irrelevant feature $\dot{y}_e^2$. To facilitate the circulation of the features in the channels and to reduce information redundancy. Group convolution (GC) and pointwise convolution (PC) are performed on $\dot{y}_e^1$. Specifically, GC first groups the input feature maps by channel and then applies regular convolution to each group. This approach effectively reduces the computational effort of convolution while extracting rich channel information. Unfortunately, GC prevents information exchange between different groups. To solve this problem, PC is introduced, which uses a 1×1 convolution to stitch the information from different groups to compensate for the information lost by GC. Finally, the convolved features are summed to obtain the information-rich feature map $\dot{x}_e^1$. At the same time, $\dot{y}_e^2$ is processed by PC to generate a feature map containing shallow details and concatenated with the initial input to obtain the spectral complementary feature $\dot{x}_e^2$. This stage can be formulated as

$$\begin{cases} \dot{x}_e^1 = \text{GC}\left(\dot{y}_e^1\right) + \text{PC}\left(\dot{y}_e^1\right) \\ \dot{x}_e^2 = \text{PC}\left(\dot{y}_e^2\right) \cup \dot{y}_e^2 \end{cases} \tag{5}$$

where GC$(\cdot)$ and PC$(\cdot)$ denote GC and PC, respectively. The SKNet [60] method is applied to adaptively merge the features $\dot{x}_e^1$ and $\dot{x}_e^2$. Specifically, the global average pooling (GAP) operation is used to obtain the global spectral information $\dot{x}_e^* \in R^{K \times 1 \times 1}$ with channel statistics, calculated as

$$\dot{x}_{e_m}^* = \text{GAP}\left(\dot{x}_e^m\right) = \frac{1}{I \times J} \sum_{i=1}^{I} \sum_{j=1}^{J} \dot{x}_{e,k}^m(i, j) \tag{6}$$

where $m = 1, 2$; $\dot{x}_{e,k}^m(i, j)$ is the pixel value of the feature map $\dot{x}_e^m$ at position (i, j) of channel K. The GAP$(\cdot)$ operation allows the output of each channel to be interpreted directly in terms of the confidence of the corresponding category. After that, $\dot{x}_{e_1}^*$ and $\dot{x}_{e_2}^*$ go through softmax channel soft-attention operation to get the weight vectors $s_1, s_2 \in R^K$, where $s_1 + s_2 = 1$. With the constraints of the weight vectors s_1 and s_2, the merged features by channel $\dot{x}_e^1$ and $\dot{x}_e^2$ to obtain the salient feature map $\hat{x}_e$ are as follows:

$$\hat{x}_e = s_1 \dot{x}_e^1 + s_2 \dot{x}_e^2. \tag{7}$$

C. ISSC Module (l_2–l_4)

The HSI denoising problem is a typical inverse problem, and its solution exists ill-posed. Moreover, current supervised denoising methods usually establish constraints directly on the ground truth and the denoised image, which cannot predict the intermediate feature changes, and the model is not stable and prone to multiple solutions. Therefore, we design an ISSC module, which supervises intermediate features by inverse

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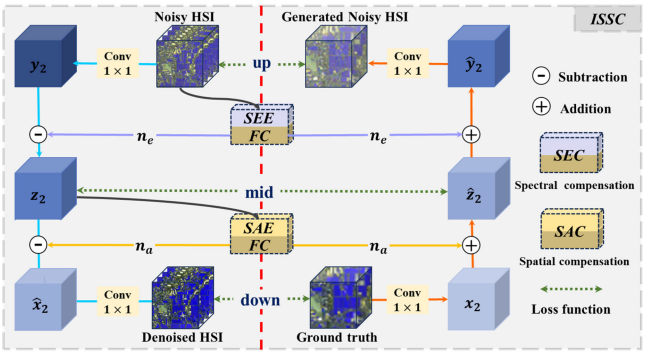


Fig. 5. ISSC module. The left of the red line is the forward denoising process and the right side is the inverse additive noise process. SEC-block and SAC-block are included.

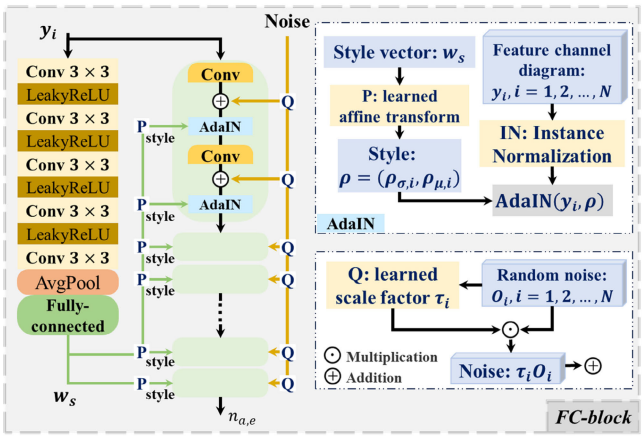


Fig. 6. FC-block. The mapping network f consists of multilayer convolution and the synthesis network g consists of nine-layer stylization modules. "P" stands for the learned affine transformation and "Q" applies the learned per-channel scaling factor to the noisy input.

constraints, facilitates the circulation of multiscale feature information, and improves model stability. Meanwhile, style transfer [61] is introduced into the network and is used to design spectral compensation block (SEC-block) and spatial compensation block (SAC-block), to accurately compensate for the lost spatial and spectral detail features using superior fine feature control ability.

In MSIC-Net, the degraded HSI has the vast majority of the noise removed when passing through the SSFI module, but the small-scale denoising mainly focuses on the local features, which may not be able to accurately retain the global structure and texture information of the image. Therefore, the designed ISSC module is applied to l_2–l_4, which is capable of extracting residual noise from different scales while considering the global information and realizing the information compensation. Fig. 5 illustrates the denoising process of the second layer (l_2) ISSC module. The ISSC module contains two main parts: the forward denoising process and the inverse additive noise process. In the forward denoising process, the input noisy image y_2 passes through the SEC-block to get the noise n_e with spectral distribution characteristics, which is removed to get the intermediate feature z_2; then, the intermediate feature passes through the SAC-block to get the noise n_a with spatial distribution characteristics, which is removed to obtain the predicted clean image $\hat{x}_2$. It is considered that when the extracted noise is sufficiently accurate, the ground truth is summed with the noise signal to obtain the original noisy HSI image. Specifically, the clean reference image x_2 is summed with the noise n_a to obtain the intermediate feature $\hat{z}_2$, which is highly consistent with z_2. Similarly, the addition of $\hat{z}_2$ to the noise n_e yields the noisy image $\hat{y}_2$, which is highly consistent with y_2. The MSIC-Net network relies on the advantage of parameter sharing between different layers of the ISSC module, which moreover improves the generalization ability.

To realize the compensation of spatial and spectral information, SAC-block and SEC-block are designed. In MSIC-Net, we use the SEE unit and SAE unit to extract noise features with spectral/spatial distribution characteristics from both the spectral and spatial dimensions and then rely on FC-block to refine the noise features. FC-block, as shown in Fig. 6, has the main advantages of enabling fine control of the noise style, and improving the diversity and quality of the extracted noise through random noise injection and multi-scale training strategies. Specifically, the noise vector w with spatial/spectral distribution characteristics is passed through a nonlinear mapping network $f: w \rightarrow w_s$ to generate the style vector w_s, which is used to control the different levels of detail information of the generated noise so that the network can better capture the complex detail characteristics, while the mapping network f consists of five layers of 3×3 convolutional layers, an average pooling (AvgPool) layer, and fully connected layers. Subsequently, the style vector w_s undergoes a learnable affine transformation P to generate the scaling factor and deviation factor $\rho = (\rho_\sigma, \rho_\mu)$. These two factors are weighted and summed with the output after instance normalization (IN) to achieve the overall style control of the noise in the synthesis network g. The specific implementation of AdaIN is as follows:

$$\text{AdaIN}(y_i, \rho) = \rho_{\sigma,i} \frac{y_i - \mu(y_i)}{\sigma(y_i)} + \rho_{\mu,i}. \tag{8}$$

Among them, the synthesis network g consists of nine stylization modules, each with a resolution of 32×32 and two AdaIN layers, which mainly control the detailed features of the generated noise. In addition, after the convolutional layer of each stylization module and before the AdaIN layer, random noise at the level of the channel feature maps is added, and the noise inputs of each channel in each layer are shared, but they need to be multiplied by the learned weights before being added to the feature maps, as shown in the following equation:

$$y_n = y + \tau_Q n \tag{9}$$

where y_n is the image after adding noise, τ_Q is the learned weight to control the intensity of the noise, and n is the input random noise. The addition of random noise allows for random control of the more subtle generation results of the noise, accurate extraction of the noise remaining in the spatial and spectral, and the realization of the compensation of spatial–spectral information.

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D. Loss Function

The MSIC-Net is optimized by end-to-end training. To consider the balance between model stability and denoising performance, we design a multiconnection loss based on the mean squared error (mse) with inverse constraints to train the proposed network. The overall denoising loss is designed as follows:

$$\mathcal{L}_{\text{all}} = \mathcal{L}_{\text{stage}} + \alpha_1 \mathcal{L}_{\text{stage}_1} + \alpha_2 \mathcal{L}_{\text{stage}_2} + \alpha_3 \mathcal{L}_{\text{stage}_3} + \alpha_4 \mathcal{L}_{\text{stage}_4} \tag{10}$$

where $\mathcal{L}_{\text{stage}}$ is the MSIC-Net output image $\hat{X}$ and the reference image X; $\mathcal{L}_{\text{stage}_i}, i = 1, 2, 3, 4$, denotes the loss function of different scales; and α_i represents the weight coefficients. When $i = 1$, $\mathcal{L}_{\text{stage}_1}$ is the loss function of the degraded image to get the image $\hat{x}_1$ after the SSFI module and the reference image X to get the image x_1 after downsampling. When $i = 2$, $\mathcal{L}_{\text{stage}_2}$ is the loss calculated by the noisy image through the l_2, based on the ISSC module, the formula of $\mathcal{L}_{\text{stage}_2}$ is as follows:

$$\mathcal{L}_{\text{stage}_2} = \mathcal{L}_{\text{up}_2} + \mathcal{L}_{\text{mid}_2} + \mathcal{L}_{\text{down}_2} \tag{11}$$

where $\mathcal{L}_{\text{up}_2}$ denotes the second scale input noisy image y_2 and the generated noise-containing image $\hat{y}_2$, $\mathcal{L}_{\text{mid}_2}$ denotes the intermediate feature image, and $\mathcal{L}_{\text{down}_2}$ denotes the output clean image $\hat{x}_2$ and the reference image X after downsampling to get the image x_2 (as shown in Fig. 5), $i = 3, 4$ same as above. The multiconnection loss allows the model to achieve better denoising effect at different scales and improves the robustness of the model.

IV. Experimental Analysis

In this section, we present the relevant experimental setup in Section IV-A. Subsequently, simulated and real HSI denoising experiments are revealed in Sections IV-B and IV-C. To further validate the effectiveness, classification experiments are conducted in Section IV-D. Finally, ablation experiments are performed on the components in MSIC-Net in Section IV-E.

A. Experimental Setting

For the dataset of the simulation experiment, we processed the HSIs from ZY1-02D satellite to obtain about 16 000 clean–noise pairs as the training set and about 700 pairs as the validation set. Specifically, mixed noise is added to the ground truth, mainly including the following six cases.

1) Case 1 (Gaussian Noise): We added zero-mean Gaussian noise to each band of the HSI. The signal-to-noise ratio of the Gaussian noise was set to $G = [5–30 \text{ dB}]$.

2) Case 2 (Gaussian Noise + Impulse Noise): Based on Case 1, impulse noise was added to each band of the HSI for noise pollution, where the density of the impulse noise was set to $I = [0.010–0.015]$.

3) Case 3 (Gaussian Noise + Impulse Noise + Stripes): The case is based on Case 2 where one-third of the bands are randomly selected to be corrupted by stripes, where the ratio of the number of stripes to the total number of columns is set to $S = [5\%–30\%]$.

4) Case 4 (Gaussian Noise + Impulse Noise + Deadlines): The case is based on Case 2 where one-third of the bands are randomly selected to be corrupted by deadlines, where the ratio of the number of deadlines to the total number of columns is set to $D = [5\%–30\%]$.

5) Case 5 (Gaussian Noise + Stripes + Deadlines): In addition to adding Gaussian noise in each band of the HSI, one-third of the bands were randomly selected to add stripes and deadlines, where the numbers of stripes and deadlines as a percentage of the total columns were the same as in Cases 3 and 4.

6) Case 6 (Gaussian Noise + Impulse Noise + Stripes + Deadlines): Impulse noise of the same density as in Case 2 was added to all bands, based on Case 5.

All deep learning-based methods were trained end-to-end on NVIDIA GeForce RTX 3080 Ti based on the above six cases, where the patch size of the training is set to $128 \times 128 \times 25$, the batch is 4, and the number of epochs is 200. The MSIC-Net implemented by the PyTorch framework is updated using the adaptive moment estimation optimizer, which employs a stepwise learning rate decay strategy, where the learning rate is initialized to 10^{-3} and kept constant for the first 80% of the epochs; for the remaining 20% of the epochs, the learning rate is multiplied by 0.1 for each epoch that passes. The realization quickly approximates the optimal solution in the early epochs of training, while the parameters are meticulously adjusted in the later stages of training to better approximate.

To verify the reliability of the MSIC-Net, ten HSI denoising algorithms are chosen as comparison methods. These include model-driven methods: BM4D [4] based on nonlocal similarity, MTGSNMF [5] based on sparse representation, LRTV [20] and FastHyMix [23] based on low rank, and LRTDTV [34] based on tensor decomposition. Data-driven methods for joint spatial–spectral denoising are HSID-CNN [9], MAC-Net [51], SQAD [41], NSSNN [39], and SMDS-Net [56].

In addition, quantitative and qualitative evaluations are used to analyze the denoising results. For quantitative evaluation, we selected six quality metrics: correlation coefficient (CC) [62], spectral angle mapper (SAM) [63], erreur relative globale adimensionnelle de synthèse (ERGAS) [64], structure similarity index measure (SSIM) [65], root-mean-square error (RMSE) [66], and peak signal-to-noise ratio (PSNR) [67]. Generally, CC and SSIM nearing 1, higher PSNR, and smaller SAM, ERGAS, and RMSE indicate better denoising quality of HSI. For qualitative evaluation, we perform a comparison of the denoised HSI with the reference image visually, focusing on its restored sharpness and local details. The quantitative and qualitative evaluations can help us to evaluate the HSI denoising methods comprehensively.

B. Simulation Experiment Results

In the experiments, the Xiong'an dataset [68] was chosen as the simulation data. The dataset contains 250 bands in the spectral range of 400–1000 nm, and the Xiong'an data tested in the experiment are cropped to a shape of $128 \times 128 \times 225$. To test the stability of the denoising method for different noises, we performed experiments on the most severely noisy simulations of the six cases.

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TABLE I

QUANTITATIVE COMPARISON OF DIFFERENT METHODS FOR XIONG'AN DATASET IN CASES 1–6. THE BEST VALUE IS MARKED IN BOLD AND THE SECOND VALUE IS UNDERLINED

Index	BM4D	MTGSNMF	LRTV	FastHyMix	LRTDTV	HSID-CNN	MAC-Net	SQAD	NSSNN	SMDS-Net	MSIC-Net
Case 1: Gaussian noise (5dB)											
CC	0.9705	0.9620	0.9710	**0.9933**	0.9833	0.9255	0.8987	0.9868	0.9814	0.9802	0.9908
SAM	4.2279	8.8594	6.3769	2.0478	3.0985	9.0939	10.0848	3.1683	8.9179	3.0490	**2.0229**
ERGAS	8.7236	26.1595	10.5979	6.0913	9.0150	17.6483	27.2357	9.3002	14.6697	6.6056	**5.4379**
SSIM	0.8602	0.7727	0.8762	**0.9598**	0.9093	0.8576	0.8189	0.8857	0.8774	0.9273	0.9594
RMSE	0.0272	0.0706	0.0361	0.0145	0.0254	0.0450	0.0537	0.0283	0.1262	0.0213	**0.0127**
PSNR	30.0337	25.4708	32.5627	32.8783	31.1587	28.1522	24.3441	30.5202	29.9165	32.8780	**33.1359**
Case 2: Gaussian noise (5dB) + Impulse noise (0.015)											
CC	0.8204	0.9828	0.9686	0.9726	**0.9849**	0.9636	0.9798	0.9521	0.9747	0.9713	0.9830
SAM	9.1671	3.2678	6.7684	2.5694	2.8427	5.7031	3.8072	4.6354	3.9500	4.0354	**2.1868**
ERGAS	61.4995	16.1120	23.3502	8.0422	8.8519	17.8235	8.8502	9.4937	8.6920	8.5128	**7.0275**
SSIM	0.6893	0.9045	0.8581	0.9163	0.9039	0.8239	0.8936	0.8777	0.8825	0.8836	**0.9216**
RMSE	0.1127	0.0378	0.0652	0.0252	0.0302	0.0433	0.0242	0.0347	0.0338	0.0227	**0.0212**
PSNR	17.1088	27.2373	25.1830	33.0096	30.7090	28.3268	32.1678	31.3066	31.5287	**33.0932**	32.8656
Case 3: Gaussian noise (5dB) + Impulse noise (0.015) + Stripes (30%)											
CC	0.8171	0.9266	0.9729	0.9651	0.9781	0.9705	0.9805	0.9674	0.9629	0.9664	**0.9810**
SAM	9.4957	6.1994	5.2259	5.5606	4.0319	5.0044	**3.3901**	20.5724	4.6449	4.3127	2.856
ERGAS	61.8436	18.931	16.5079	17.7439	9.1284	17.4198	**7.8428**	33.6491	11.0947	9.9584	8.1109
SSIM	0.6694	0.7883	0.8750	0.8645	0.8884	0.8598	**0.9162**	0.7508	0.8646	0.8715	0.9130
RMSE	0.1134	0.0508	0.0487	0.0418	0.0257	0.0445	0.0232	0.1198	0.0312	0.0247	**0.0215**
PSNR	17.0566	25.6941	28.1418	29.4118	32.5090	28.0021	33.5549	29.6385	29.1031	32.1743	**33.8478**
Case 4: Gaussian noise (5dB) + Impulse noise (0.015) + Deadlines (30%)											
CC	0.7483	0.8609	0.9620	0.9491	0.9764	0.9578	0.9756	0.9437	0.9694	0.9629	**0.9796**
SAM	13.4686	11.0272	8.1342	8.4824	4.4742	6.5248	5.8236	7.1829	6.0942	6.6066	**3.9124**
ERGAS	72.2747	24.1029	32.722	24.9200	10.1444	22.9508	13.1063	16.1056	14.0262	17.6869	**9.0753**
SSIM	0.6046	0.7270	0.8364	0.8110	0.8855	0.7976	0.8904	0.8382	0.8882	0.8578	**0.9092**
RMSE	0.1289	0.0662	0.0799	0.0669	0.0301	0.0627	0.0351	0.0522	0.0422	0.0427	**0.0260**
PSNR	16.8104	30.7125	24.5222	25.7002	32.6897	25.5414	31.8582	29.1294	31.3710	31.8472	**32.7330**
Case 5: Gaussian noise (5dB) + Stripes (30%) + Deadlines (30%)											
CC	0.8456	0.8460	0.9625	0.9519	0.9725	0.7784	0.9080	0.9633	0.92981	0.9663	**0.9728**
SAM	17.4445	17.6816	16.7041	15.7681	12.3401	14.2767	24.8762	20.6483	15.8749	12.8586	**11.4703**
ERGAS	33.1877	37.4763	39.7954	32.8492	30.4495	32.6191	42.3358	33.8142	30.2973	30.7016	**29.7486**
SSIM	0.6271	0.5997	0.6917	0.7510	0.8283	0.4995	0.6311	0.7337	0.7265	0.8496	**0.8511**
RMSE	0.1157	0.1423	0.1583	0.1236	0.1111	0.0975	0.1969	0.1199	0.1097	0.0910	**0.08644**
PSNR	26.0617	23.7912	23.1398	25.4758	30.2347	22.3431	24.9874	29.6323	25.6881	30.9484	**31.3204**
Case 6: Gaussian noise (5dB) + Impulse noise (0.015) + Stripes (30%) + Deadlines (30%)											
CC	0.7240	0.80373	0.9594	0.9358	0.9745	0.9600	0.8018	0.9322	0.9266	0.9579	**0.9753**
SAM	14.8428	13.6633	8.8217	9.6913	4.2859	7.2740	20.5308	6.7609	8.2534	7.4425	**4.7330**
ERGAS	71.3855	34.3577	41.7984	28.7852	10.4656	26.9730	51.9841	25.5053	21.034	19.9691	**9.9147**
SSIM	0.5667	0.62322	0.8107	0.7615	0.8932	0.7964	0.5191	0.7767	0.7914	0.8514	**0.8751**
RMSE	0.1296	0.122674	0.0963	0.0838	0.0309	0.0797	0.2843	0.1336	0.05468	0.0498	**0.0285**
PSNR	17.0369	26.0504	23.1493	24.2666	31.9958	23.8507	21.9058	31.3139	28.6492	31.1270	**32.6130**

In Table I, the quantitative result values of 11 methods under Cases 1–6 are shown. Compared with other methods, our method obtains comparative denoising results for spectral and spatial recovery. BM4D and LRTV are more effective for Gaussian noise denoising but less so than the other methods for mixed noises. FastHyMix effectively removes both Gaussian and impulse noise and is less capable for stripes and deadlines. LRTDTV and SMDS-Net are more stable for different complex noise types, but their denoising effect is weaker in Case 5, while MSIC-Net is relatively more stable in denoising performance for different complex noise types. For qualitative evaluation comparison, the pseudo-color denoising results for the Xiong'an dataset with (10, 20, 30) in Case 1 are shown. To demonstrate the quality of the recovery, we provide images

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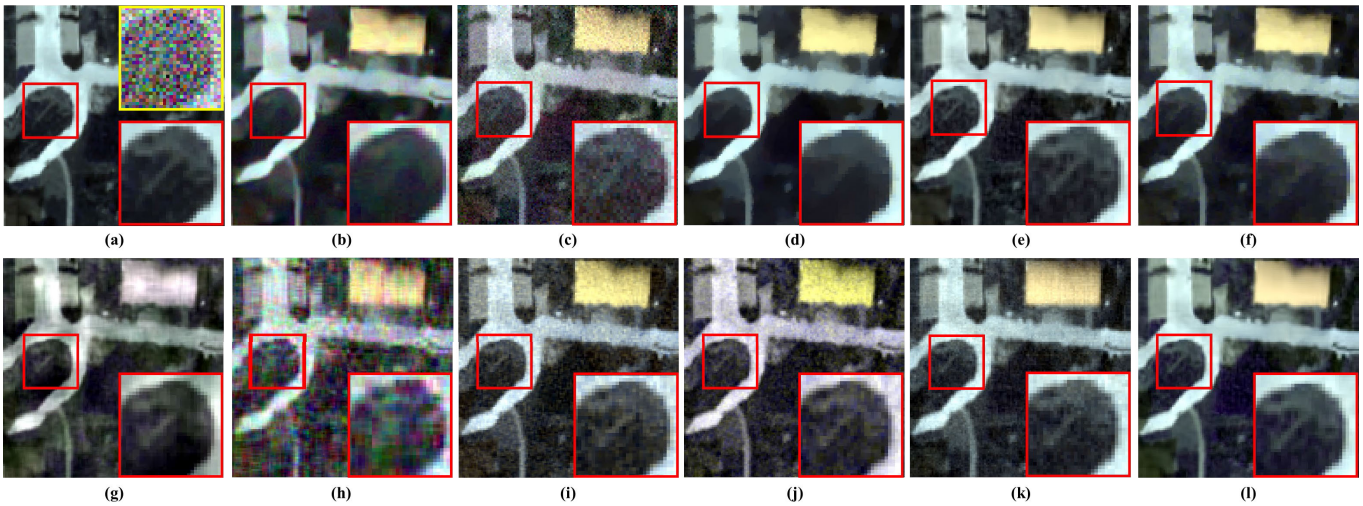


Fig. 7. Case 1 qualitative comparison on the Xiong'an dataset with (10, 20, 30). (a) Clear. (b) BM4D. (c) MTSNMF. (d) LRTV. (e) FastHyMix. (f) LRTDTV. (g) HSID-CNN. (h) MAC-Net. (i) SQAD. (j) NSSNN. (k) SMDS-Net. (l) MSIC-Net.

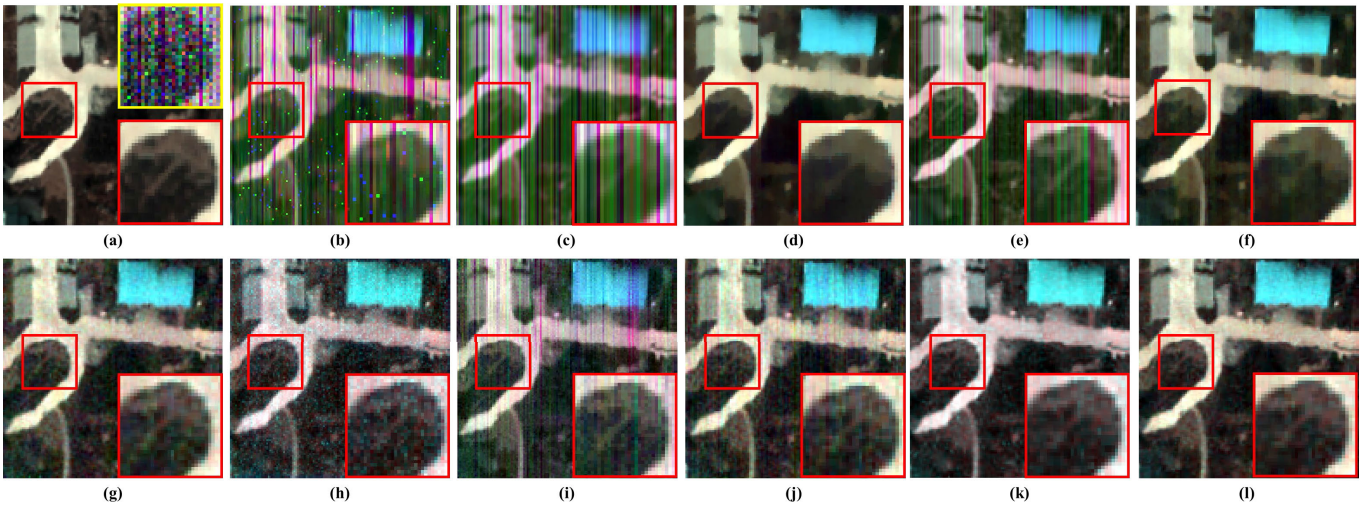


Fig. 8. Case 6 qualitative comparison on the Xiong'an dataset with (70, 19, 10). (a) Clear. (b) BM4D. (c) MTSNMF. (d) LRTV. (e) FastHyMix. (f) LRTDTV. (g) HSID-CNN. (h) MAC-Net. (i) SQAD. (j) NSSNN. (k) SMDS-Net. (l) MSIC-Net.

of local details. In Fig. 7, the model-driven methods rely on explicit mathematical models and a priori knowledge, and their denoising performance is relatively more stable than that of the data-driven methods when dealing with Gaussian noise. In addition, we show the pseudo-color denoising results for the simulated dataset bands (70, 19, 10) in Case 6 in Fig. 8, where the data-driven methods rely on a strong nonlinear fitting capability and perform well in dealing with complex mixed noise. As can be seen from the above figures, LRTV and LRTDTV perform well in the model-driven methods and successfully remove most of the noise. However, the TV regularization leads to an oversmoothing of the image, resulting in the loss of some detail information. The BM4D removes only Gaussian noise, while MTSNMF and FastHyMix can remove Gaussian and impulse noise, but there are still obvious stripes and deadlines. In all the data-driven methods, although the spatial and spectral information of the HSI is taken into account, there is insufficient compensation for the loss of information in the denoising process. HSID-CNN produces blurred results, and MAC-Net loses the detailed information of the image. SQAD is unable to completely remove all the noise, NSSNN removes most of the noise, but there are obvious artifacts in some areas. SMDS-Net removes noise better, but there is more significant spatial distortion. In contrast, our MSIC-Net effectively removes noise and better preserves details through ISSC operation.

To evaluate the spectral recovery quality of each method, Fig. 9 shows the SAM after denoising for Case 6. The color from blue to red in the figure indicates the gradual increase of spectral distortion, and the results show that our MSIC-Net can recover the spectral information better. In addition, we perform a linear regression analysis on the reference HSI and the denoised image in Case 1 and generate scatter density plots to visualize the results. In the unary linear regression analysis, the coefficient of determination (R^2) is approximately equal to the square of the CC. This indicates that if the CC between the reference image and the recovered image is high, there is a strong linear relationship and thus a high level of fit

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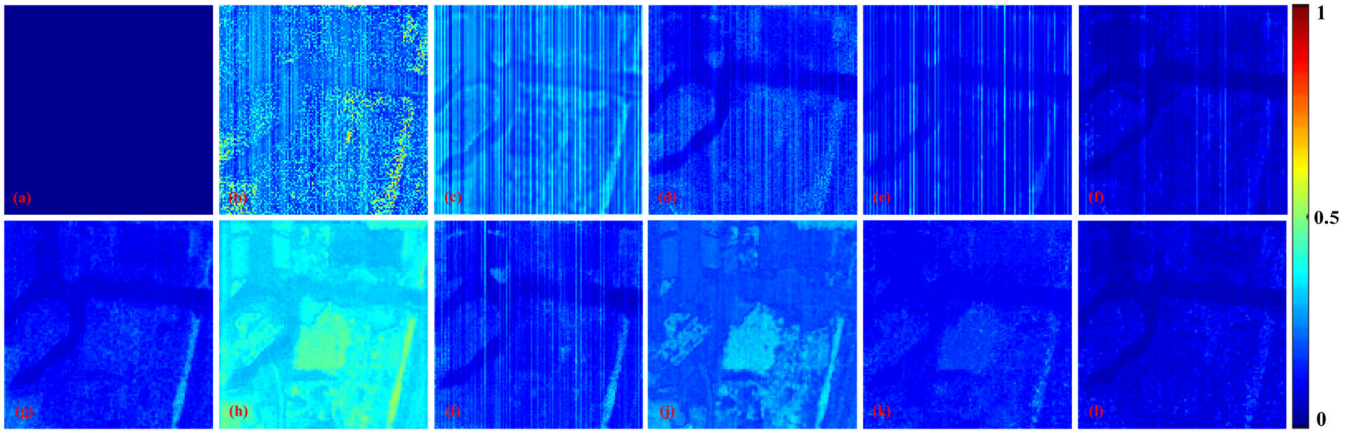


Fig. 9. Spectral angle mapping of denoising methods in Case 6. (a) Clear. (b) BM4D. (c) MTSNMF. (d) LRTV. (e) FastHyMix. (f) LRTDTV. (g) HSID-CNN. (h) MAC-Net. (i) SQAD. (j) NSSNN. (k) SMDS-Net. (l) MSIC-Net.

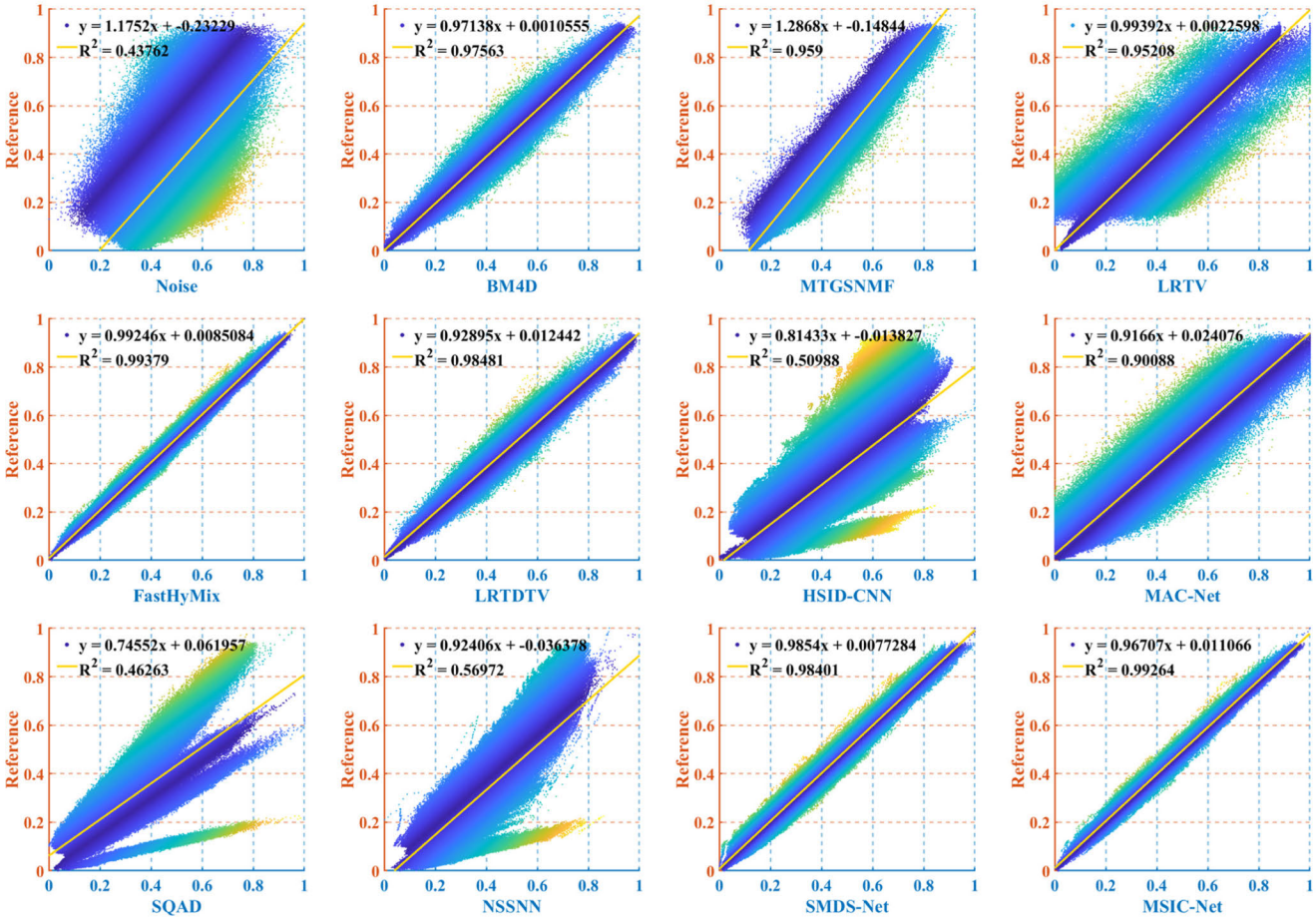


Fig. 10. Scatter density plots for linear regression analysis of reference and denoised images in Case 1.

linear regression model. Combined with the spatial results in Fig. 7, some of the methods in Fig. 10 show better recovery of spatial information, but the spectral information fitting is still deficient. In contrast, FastHyMix, SMDS-Net, and our MSIC-Net perform well in fitting the reference image, demonstrating stable spatial and spectral recovery.

To evaluate the recovery results of different methods, we plot the band-wise PSNR values for the Xiong'an dataset, as shown in Fig. 11. All the methods can remove the noise to some extent. However, in most of the bands, the PSNR values of MSIC-Net are higher than those of the other methods. Especially when the image is corrupted by stripes and deadlines, our method achieves more stable denoising results.

C. Real Experiment Results

To further demonstrate the reliability and adaptability of the proposed MSIC-Net, the denoising ability of MSIC-Net is compared with other HSI algorithms on two real HSI datasets: the PRISMA and Indian Pines. To effectively evaluate the

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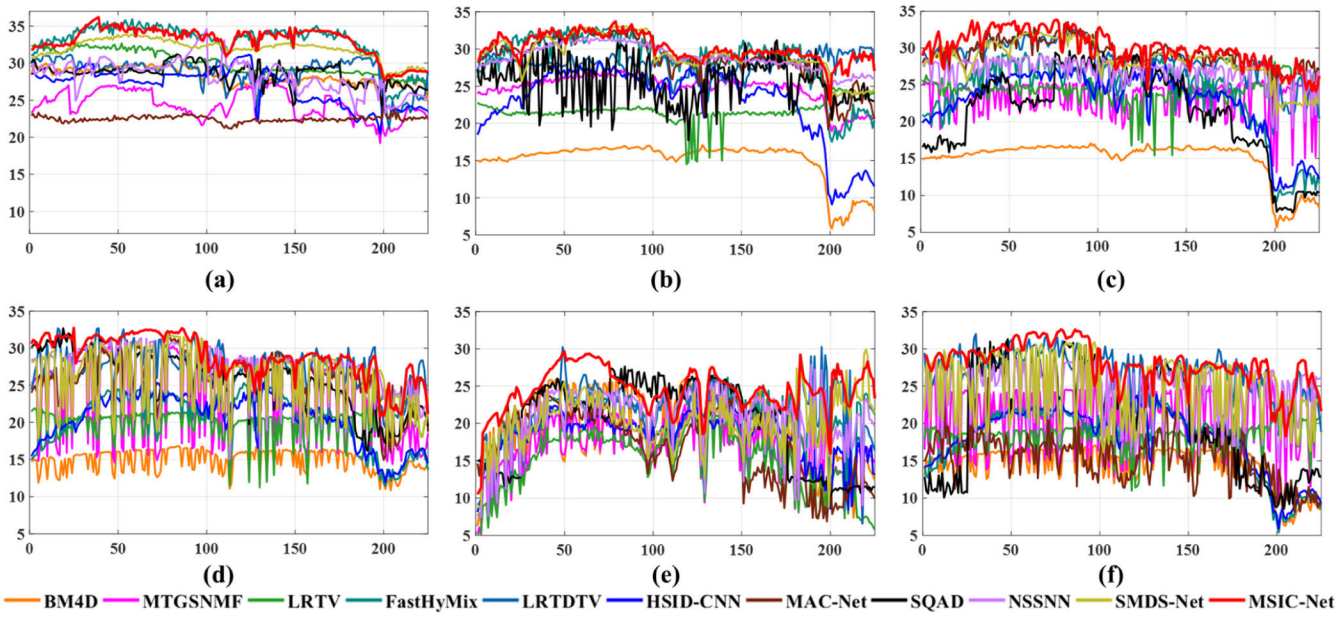


Fig. 11. (a)–(f) Case 1–6 PSNR values for the Xiong'an dataset.

TABLE II
REAL HSI DENOISING RUNTIMES

Time/s	BM4D	MTGSNMF	LRTV	FastHyMix	LRTDTV	HSID-CNN	MAC-Net	SQAD	NSSNN	SMDS-Net	MSIC-Net
PRISMA	166.9467	281.6328	22.7540	4.3156	64.1820	14.7216	28.5367	19.8487	25.5458	19.7074	17.9980
Indian Pines	186.9803	301.3470	24.3468	5.7541	73.8094	15.6688	30.5612	22.0602	28.5107	25.4732	19.2169

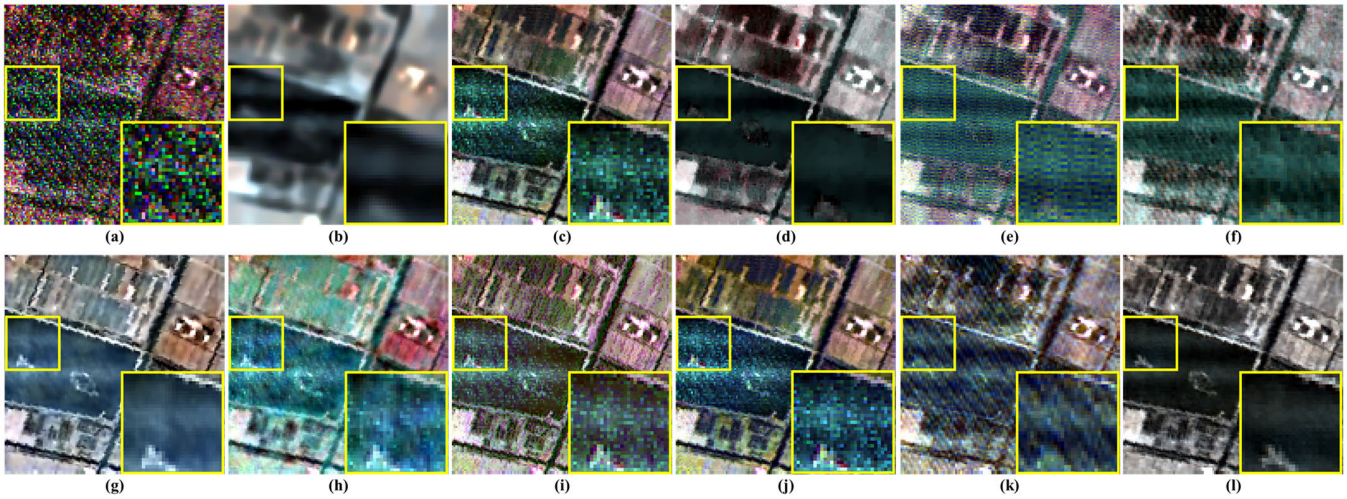


Fig. 12. Denoising results for the PRISMA dataset (3, 2, 1). (a) Noise HSI. (b) BM4D. (c) MTSNMF. (d) LRTV. (e) FastHyMix. (f) LRTDTV. (g) HSID-CNN. (h) MAC-Net. (i) SQAD. (j) NSSNN. (k) SMDS-Net. (l) MSIC-Net.

computational efficiency, the running time of methods is also shown in Table II. The specific results and analysis are given as follows.

1) PRISMA Dataset (128 × 128 × 150): This dataset is corrupted by Gaussian noise, impulse noise, stripes, and deadlines, as shown in Fig. 12(a), the pseudo-color is composed of (3, 2, 1). From Fig. 12(b) to (l), BM4D removes noise but loses a lot of information. MTSNMF, FastHyMix, MAC-Net, NSSNN, and SMDS-Net remove most of the noise, but impulse noise remains in the denoising result. Although LRTV and LRTDTV remove noise, much detailed information is smoothed. SQAD eliminates noise and produces a large number of spatial artifacts. In contrast, MSIC-Net performs best, effectively removing noise and achieving high spatial–spectral fidelity.

2) Indian Pines Dataset (128 × 128 × 200): This dataset is corrupted by Gaussian and impulse noise, as shown in Fig. 13(a), with pseudo-color results of (1, 2, 199). As shown in Fig. 13(b)–(l), BM4D and FastHyMix can reduce some noise, but dense impulse noise remains in the restoration

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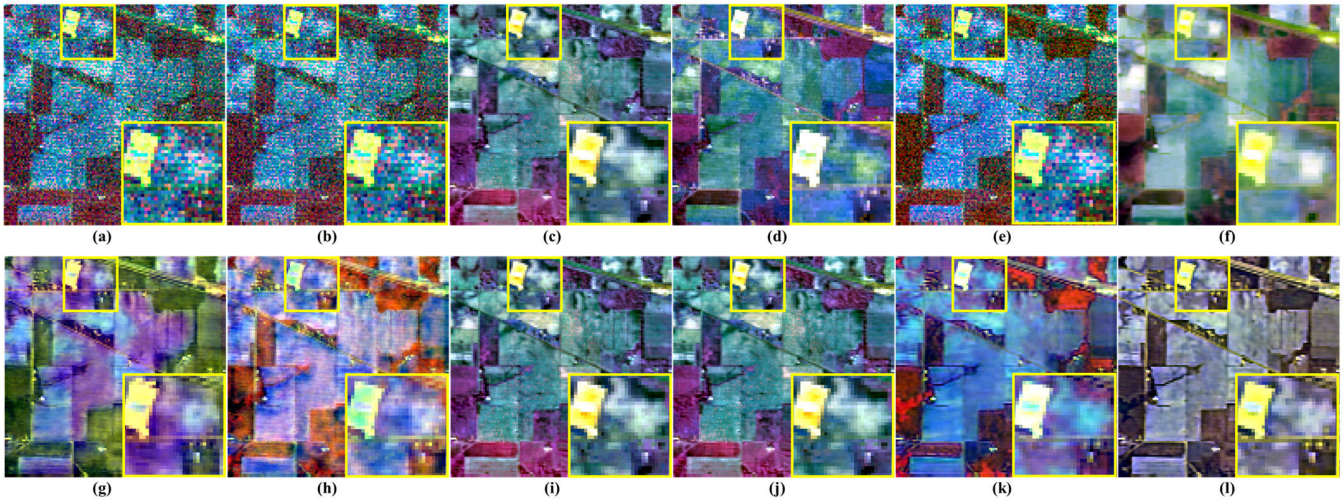


Fig. 13. Denoising results for the Indian Pines dataset (1, 2, 199). (a) Noise HSI. (b) BM4D. (c) MTSNMF. (d) LRTV. (e) FastHyMix. (f) LRTDTV. (g) HSID-CNN. (h) MAC-Net. (i) SQAD. (j) NSSNN. (k) SMDS-Net. (l) MSIC-Net.

TABLE III
INDIAN PINES CLASSIFICATION ACCURACY RESULTS

Classification	Noise HSI	BM4D	MTGSNMF	LRTV	FastHyMix	LRTDTV	HSID-CNN	MAC-Net	SQAD	NSSNN	SMDS-Net	MSIC-Net
OA	75.28%	77.37%	77.26%	78.46%	80.21%	82.35%	84.27%	80.60%	82.21%	83.64%	84.72%	**85.24%**
Kappa	0.7088	0.7339	0.7325	0.7516	0.7529	0.8118	0.8338	0.7884	0.8230	0.8280	0.8390	**0.8433**

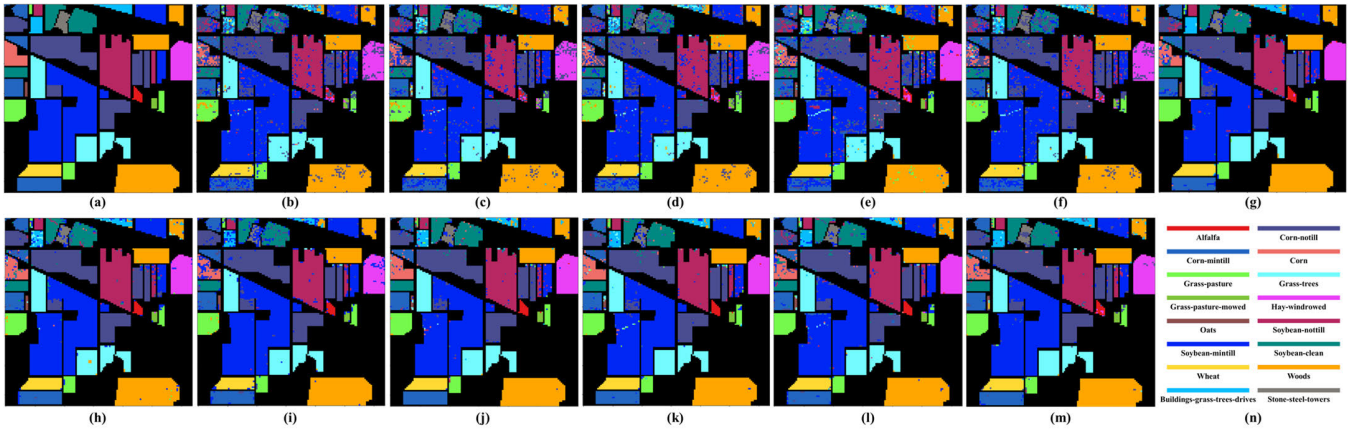


Fig. 14. SVM classification results for the Indian Pines dataset. (a) Ground truth. (b) Original. (c) BM4D. (d) MTSNMF. (e) LRTV. (f) FastHyMix. (g) LRTDTV. (h) HSID-CNN. (i) MAC-Net. (j) SQAD. (k) NSSNN. (l) SMDS-Net. (m) MSIC-Net. (n) 16 classes.

results. The LRTDTV method also shows the ability of noise suppression; however, some detailed information is smoothed. Other methods effectively suppressed the noise, and the recovery results showed significant spatial–spectral distortion. MSIC-Net performs best, removing noise while preserving detailed information.

Combining Table II runtimes with the results visualized in Figs. 12 and 13, it is observed that although MSIC-Net is not the fastest in terms of computation time, the method excels in high-quality denoising and provides an effective solution to trade off computational cost and performance.

D. Classification Experiment

To further confirm the efficacy of the proposed method, we conducted classification experiments on the denoising results of the Indian Pines dataset. The dataset contains 16 categories and the training set consists of 30% randomly generated samples from each category. The denoising results of all methods were tested for classification accuracy using a support vector machine [69]. The classification results are shown in Fig. 14 and the quantitative results for overall accuracy (OA) and kappa coefficient are listed in Table III. Before denoising, as shown in Fig. 14(b), the classification results show discontinuities, with the OA and kappa coefficient being only 75.28% and 0.7088, respectively. However, after denoising, as shown in Fig. 14(c)–(m), the OA and kappa coefficient improve significantly. However, methods in Fig. 14(c)–(g) still show significant fragmentation in the classification results due to poor denoising. Methods in Fig. 14(h)–(m) with excellent denoising performance demonstrate satisfactory classification

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TABLE IV
ISSC MODULE ABLATION EXPERIMENT QUANTITATIVE RESULTS

ISSC	ERGAS	RMSE	MPSNR
√	**9.9147**	**0.0285**	**27.8828**
×	10.4656	0.0309	27.0061

TABLE V
FC-BLOCK ABLATION EXPERIMENT QUANTITATIVE RESULTS

FC-block	ERGAS	RMSE	MPSNR
√	**9.9147**	**0.0285**	**27.8828**
×	12.1090	0.0325	26.3115

results, but method in Fig. 14(m) performs best in detail. The classification accuracy results in Table III further confirm the effectiveness of the proposed MSIC-Net, which obtains the highest OA and kappa coefficient values of 85.24% and 0.8433, respectively.

E. Ablation Experiment

To validate the effectiveness of the modules in MSIC-Net, a comprehensive ablation study was conducted. The following ablation experiments were validated under Case 6.

1) ISSC Module: In this experiment, we test the effectiveness of the ISSC module and Table IV shows the ablation results. The module imposes supervisory constraints on the intermediate variables by introducing inverse constraints and combines with the multiconnection loss to realize multiscale information flow, which improves the stability and robustness of the model. As a result, the denoising network with inverse constraints significantly improves the denoising performance compared to the denoising network without inverse constraints, with a reduction of 0.55 in ERGAS, a reduction of 0.024 in RMSE, and an improvement of 0.88 in mean PSNR (MPSNR).

2) Feature Compensation Block: This ablation experiment investigates the effect of the ISSC module subcomponent FC-block on the network. The FC-block with style transfer in the network is to accurately extract the residual noise. In Table V, it can be seen that the denoising network with FC-block reduces ERGAS by 2.1943, RMSE by 0.004, and MPSNR gain by 1.5713 compared to the denoising network without FC-block, which indicates that FC-block is important for the compensation of spatial and spectral information.

3) SEC/SAC-Block: The degraded HSI has extracted most of the noise after the SSFI module, but there is still residual noise existing in both spatial and spectral dimensions. Therefore, in the ISSC module, we designed a dual branch of the spatial and spectral compensation module to extract the residual noise to minimize the information loss. To verify the importance of the dual-branch compensation information, we performed ablation experiments on only the SEC-block branch and only the SAC-block branch. As can be seen from Table VI, the denoising performance of the network is higher in the presence of dual-branch compensation blocks, whereas the networks with only SAC or SEC blocks lose the corresponding information due to the lack of spectral or spatial compensation.

TABLE VI
SAC/SEC-BLOCK ABLATION EXPERIMENT QUANTITATIVE RESULTS

SAC/SEC	ERGAS	RMSE	MPSNR
SAC + SEC	**9.9147**	**0.0285**	**27.8828**
SAC	10.0239	0.0289	27.6667
SEC	10.0309	0.0291	27.6613

V. CONCLUSION

In this article, an MSIC-Net for HSI denoising is proposed. The designed SSFI module with a dual-branch structure can accentuate salient features and improve the accuracy of the feature extraction. In addition, the ISSC module with inverse constraints combined with multiconnection loss can alleviate the ill-posed problem of the denoising model and enhance the stability. The FC-block with style transfer can accurately compensate for the lost spatial and spectral details and further optimize the structure of HSI. After a number of experimental validations, MSIC-Net demonstrates excellent performance in both qualitative and quantitative evaluations, outperforming other state-of-the-art denoising methods.

Future research will focus on integrating prior theories (e.g., low-rank and sparse representation) into network architectures with feature compensation to further enhance the denoising performance and robustness of the model. In addition, efforts will be made to optimize the computational efficiency of the model to achieve a more effective denoising process.

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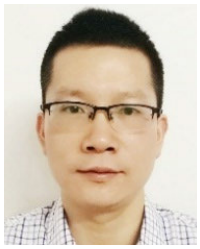
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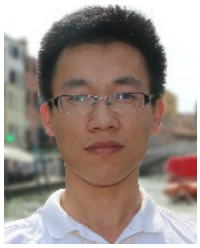
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